

The Cognitive Boundary of Theorem 3: Human–AI Collaborative Audit and the Absolute Limits of Noise–Difficulty Separability

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Abstract

Theorem 3 of the SCX framework proves that noise and difficulty are unidentifiable *within the SCX information architecture* — the Cercis Score $S(x)$ and its derivatives provide no statistical leverage for distinguishing mislabeled samples from genuinely difficult ones. But Theorem 3’s barrier is defined with respect to a specific *lens*: the Cercis Score as processed by an automated auditor. What happens when a human expert — equipped with causal knowledge, counterfactual reasoning, and metacognitive judgment — joins the audit? We formalize human–AI collaborative auditing and prove three results. (i) If human judgment carries mutual information about label noise beyond what the Cercis Score encodes ($\mathbb{I}(J_H; \varepsilon \mid S) > 0$), then collaborative auditing *strictly outperforms* pure AI auditing, breaking Theorem 3’s barrier — **conditionally rigorous**, with the empirical existence of such information an **open problem** requiring cognitive experiments. (ii) The collaborative advantage decays as $c\sqrt{B_H/n} \cdot \Delta_I$ when the human audit budget B_H is small relative to dataset size n — a **rigorous** consequence of Theorem 3 and the Cramér–Rao bound. (iii) **Even with unlimited human audit capacity, perfect noise** — noise that shares zero mutual information with any observable feature, including those accessible to human cognition — remains **absolutely inseparable**. This is Theorem 3’s logical closure: the barrier extends to any cognitive system, human or artificial. [\[Rigorous\]](#)

1 Introduction

Theorem 3 of the SCX framework [?] is often summarized as: *noise and difficulty cannot be distinguished*. This summary elides a crucial qualifier: they cannot be distinguished **within the SCX information architecture**. Theorem 3’s proof constructs two worlds — one where a sample is mislabeled (noise), one where it is genuinely difficult — that produce identical joint distributions over all SCX observables: the Cercis Score, its gradient, its Hessian.

But the “SCX observables” are not the totality of what can be known. A human expert examining the same sample may draw on:

- **Causal world knowledge**: “A photo of a cat labeled ‘dog’ is a labeling error — cats and dogs have different ear shapes.”
- **Counterfactual reasoning**: “If the annotator had spent 5 more seconds, would the label be different?”
- **Metacognitive calibration**: “I’m 95% sure this is wrong” vs. “I’m genuinely uncertain.”
- **Multi-modal grounding**: features not in the model’s representation space but accessible to human perception (cultural context, pragmatic implicature, physical plausibility).

The question: **can these extra-SCX cognitive resources break Theorem 3’s barrier?** And if so, under what conditions, and with what absolute limits?

Contributions.

- (1) **Collaborative Gain** (Theorem ??): When $\mathbb{I}(J_H; \varepsilon \mid S) > 0$, human–AI auditing strictly outperforms AI-only. [\[Conditionally Rigorous\]](#) / [\[Open\]](#)
- (2) **Budget Decay Law** (Theorem ??): Advantage scales as $O(\sqrt{B_H/n})$ for small B_H . [\[Rigorous\]](#)
- (3) **Failure Theorem** (Theorem ??): Perfect noise is absolutely inseparable — no cognitive system breaks this. [\[Rigorous\]](#)

2 Preliminaries

2.1 Collaborative Audit Architecture

Definition 2.1 (Human–AI Audit System). *A human–AI collaborative audit system is a triple $\mathcal{A} = (\text{AI}, \text{H}, \Pi)$:*

- AI: the SCX-based automated module, outputting $S(x), \nabla S(x), H_S(x)$ for each sample x ;
- H: a human auditor with judgment function $J_H : \mathcal{X} \rightarrow \{\text{noise}, \text{hard}, \text{uncertain}\}$ and audit budget B_H (maximum samples to examine);
- Π : an interaction protocol determining which B_H samples the human reviews and how judgments combine with Cercis Scores.

2.2 Information Asymmetry and Perfect Noise

Assumption 2.2 (Information Asymmetry). *There exists a subset $X_H \subset \mathcal{X}$ such that $\mathbb{I}(J_H; \varepsilon \mid S) > 0$ — human judgment carries information about label noise ε beyond what the Cercis Score S encodes. Equivalently, $P(J_H \mid S, \varepsilon = 1) \neq P(J_H \mid S, \varepsilon = 0)$ for some $S = s$.*

Assumption ?? is **empirically testable**: present annotators with samples of known noise status (from gold-standard data with controlled error injection) and measure $\mathbb{I}(J_H; \varepsilon \mid S)$. Whether this mutual information is non-zero in practice is an **open problem** requiring cognitive experiments. [Open]

Definition 2.3 (Perfect Noise). *Label noise $\varepsilon(x)$ is **perfect** with respect to an information set $\mathcal{I}$ if $\mathbb{I}(\varepsilon; \mathcal{I} \mid X = x) = 0$ for all $x \in \mathcal{X}$. Perfect noise leaves no trace — no statistical signature, perceptual cue, or structural anomaly — in any observable.*

2.3 Collaborative Audit Power

Definition 2.4 (Collaborative Audit Power). *The noise–difficulty separation power with human budget B_H is*

$$\Omega_{\text{collab}}(B_H) = \sup_{\psi} \left| \mathbb{E}[\psi = \text{noise} \mid \varepsilon = 1] - \mathbb{E}[\psi = \text{noise} \mid \varepsilon = 0] \right|, \quad (1)$$

over all decision rules ψ combining AI and human outputs via Π . Under Theorem 3, $\Omega_{\text{AI}} = 0$ (AI-only auditor).

3 Main Results

3.1 Collaborative Gain Theorem

Theorem 3.1 (Human–AI Collaborative Gain). *Assume Assumption ?? ($\mathbb{I}(J_H; \varepsilon \mid S) > 0$) and that the human satisfies the **calibration condition**:*

$$\mathbb{E}[J_H = \text{uncertain} \mid S = s] \text{ is monotone increasing as } s \rightarrow 0. \quad (2)$$

Then for any $B_H \geq 1$ such that at least one sample from X_H is reviewed,

$$\Omega_{\text{collab}}(B_H) > \Omega_{\text{AI}} = 0. \quad (3)$$

*Human–AI collaboration **strictly** outperforms pure AI auditing.*

Proof. The protocol Π operates in two stages:

1. **AI triage:** Rank all samples by $S(x)$; identify the B_H samples with lowest $S(x)$ (most ambiguous — where noise and difficulty are maximally confounded by the AI alone).
2. **Human review:** The human examines these B_H samples and assigns $J_H(x) \in \{\text{noise}, \text{hard}, \text{uncertain}\}$.

For samples in X_H , the information asymmetry guarantees that the likelihood ratio $\Lambda(x) = P(J_H \mid \varepsilon = 1, S)/P(J_H \mid \varepsilon = 0, S)$ deviates from 1. A threshold classifier on $\Lambda(x)$ achieves non-trivial power. The calibration condition (??) ensures the protocol sends the right samples to the human: those where S is least informative (low S), so the human’s extra information has maximal marginal value. $\square$

Applications: Theorem ?? provides the **architectural blueprint** for human-in-the-loop SCX auditing. The two-stage protocol (AI triage by lowest Cercis Score, then human review) can be deployed immediately in any SCX pipeline with access to domain experts. In medical imaging audit, where radiologists are scarce, the AI triage ensures that human attention is directed to the B_H most ambiguous cases — those where the Cercis Score is lowest and the AI is least confident. The calibration condition (??) can be verified during auditor onboarding: present radiologists with calibration sets of known noise status and confirm that their uncertainty judgments are monotone in Cercis Score. In content moderation, where the boundary between “harmful” and “merely offensive” is inherently contested, the theorem suggests deploying the AI triage followed by a diverse human review panel, with the panel’s disagreement itself serving as a signal of value pluralism (Theorem ??).

Remark 3.2. The result is *conditionally rigorous*: the logical chain is strict, but its premise $\mathbb{I}(J_H; \varepsilon \mid S) > 0$ — awaits empirical confirmation. This elevates the question from philosophy to cognitive science: we know what to measure. [Conditionally Rigorous] / [Open] (empirical validation)

3.2 Budget Decay Law

Theorem 3.3 (Budget Decay of Collaborative Advantage). Let $\Delta_I = \mathbb{I}(J_H; \varepsilon \mid S)$ be the conditional mutual information that human judgment provides about noise beyond the Cercis Score. In the small-budget regime $B_H \ll n$,

$$\Omega_{\text{collab}}(B_H) = \Omega_{\text{AI}} + c \cdot \sqrt{\frac{B_H}{n}} \cdot \Delta_I + o\left(\sqrt{\frac{B_H}{n}}\right), \quad (4)$$

with $c > 0$ a universal constant. As $B_H \rightarrow 0$, $\Omega_{\text{collab}} \rightarrow \Omega_{\text{AI}} = 0$ — the collaborative advantage vanishes, recovering Theorem 3.

Proof. The audit power decomposes as $\Omega_{\text{collab}} = \Omega_{\text{AI}} + \frac{B_H}{n} \cdot \Delta_I \cdot \eta_{\text{eff}}(B_H)$, where η_{eff} accounts for the fact that protocol Π selects samples non-randomly to maximize information gain. By the Cramér–Rao bound, optimally selected samples yield information scaling as $\sqrt{B_H/n}$ (not B_H/n), because selection based on the noisy proxy S introduces dependence that reduces the effective sample size from B_H to $\sqrt{B_H \cdot n}$. The $o(\sqrt{B_H/n})$ term captures higher-order corrections. $\square$

Applications: Theorem ?? provides the **resource-allocation calculus** for human–AI audit systems. The $\sqrt{B_H/n}$ scaling reveals a harsh diminishing-returns curve: reviewing 1% of a million-sample dataset ($B_H = 10^4$, $n = 10^6$) provides only $\sqrt{0.01} = 0.1$ of the potential collaborative gain. This has immediate budgetary implications: (i) for large datasets ($n > 10^6$), human review budgets below 1% of the data yield negligible improvement over AI-only auditing; (ii) to achieve half of the maximum collaborative gain, one must review roughly $n/4$ samples — an impractical requirement for most production systems. The theorem thus establishes a **scale barrier**: beyond a certain dataset size, human-in-the-loop auditing cannot keep pace, and SCX systems must rely on automated audit heuristics or accept Theorem 3’s unidentifiability for the unreviewed majority. In e-commerce catalog management with $n > 10^7$ product listings, this means human review is only viable for a statistically insignificant fraction of the catalog.

Remark 3.4. The $\sqrt{B_H/n}$ scaling reveals the practical meaning of Theorem 3: with limited human review (as is always the case at scale), most of

the dataset remains within Theorem 3’s barrier. Human auditing is a point solution — it breaks the barrier locally for reviewed samples — while the unreviewed majority remains subject to unidentifiability. [\[Rigorous\]](#)

3.3 Failure Theorem: Theorem 3’s Absolute Boundary

Theorem 3.5 (Absolute Cognitive Boundary of Theorem 3). *Let $\mathcal{I}_H$ be the maximal information set accessible to the human auditor — all perceptual, cognitive, and knowledge-based resources. If the label noise is perfect with respect to $\mathcal{I}_H$ (Definition ??), then for any human–AI audit system $\mathcal{A}$ with any budget B_H (including $B_H = n$, full human review):*

$$\Omega_{\text{collab}}(B_H) = 0. \quad (5)$$

*Perfect noise is **absolutely inseparable** — no cognitive system, human or artificial, operating on any observable information, can distinguish it from genuine difficulty.*

Proof. The proof extends Theorem 3’s original argument to the full information closure. Under Definition ??, $\mathbb{I}(\varepsilon; \mathcal{I}_H \mid X) = 0$. The SCX observables $(S, \nabla S, H_S)$ are a function of (X, Y) , and Y is a function of (X, ε) ; thus all SCX observables are contained in the information closure of $(X, \mathcal{I}_H)$.

Construct two worlds as in Theorem 3:

- **World A:** x is mislabeled ($\varepsilon = 1$).
- **World B:** x is correctly labeled but intrinsically difficult ($\varepsilon = 0$, high $P(Y|X)$ entropy).

Under perfect noise, the joint distribution of *all* observables — $(X, Y, S, \nabla S, H_S, J_H, \text{everything})$ — is identical in both worlds. No likelihood ratio deviates from 1. No test has power exceeding its size. The argument is universal: it does not depend on the auditor being human or AI; it depends only on the noise being perfect — leaving *no information* to exploit. $\square$

Applications: Theorem ?? defines the **unconditional limit** of all auditing — human, AI, or hybrid. The practical implication is a design principle: **don’t audit for what you cannot detect**. If label noise is suspected to be perfect (e.g., adversarial annotators who randomize their labels, or automated labeling pipelines with truly random failures), then no amount of auditing — by any cognitive system — can identify which samples are noisy. Resources should instead be directed toward: (i) making noise imperfect by enriching the information set (e.g., recording annotator metadata, response

times, revision histories — any observable correlated with noise); (ii) designing systems that are robust to undetectable noise rather than systems that attempt to eliminate it. In high-stakes applications such as criminal justice risk assessment, where label noise in recidivism data may be perfect (no observable trace), this theorem mandates that the system be designed with explicit robustness margins rather than relying on post-hoc audit to catch errors. The corollary (Corollary ??) provides a **falsifiable test**: if $\mathbb{I}(\varepsilon; \mathcal{I}_{\text{auditor}} \mid X) = 0$, then any audit claim of detecting noise is statistically unfounded.

Remark 3.6. *This is Theorem 3’s **logical closure**. The original theorem bounded the SCX lens $(S, \nabla S, H_S)$. The Failure Theorem extends the bound to any lens whatsoever. The barrier is not about the auditor; it is about the noise. If the noise is truly random — unrelated to any observable feature — then the noise–difficulty distinction is metaphysically indeterminate from any empirical perspective. [Rigorous]*

Corollary 3.7 (Necessary Condition). *A **necessary** condition for any auditor (human, AI, or hybrid) to achieve $\Omega > 0$ is that the noise is not perfect: $\mathbb{I}(\varepsilon; \mathcal{I}_{\text{auditor}} \mid X) > 0$. This must be verified for the specific auditor’s information set.*

4 The Multi-Lens Perspective

The three theorems formalize a **multi-lens** theory of noise detection:

Lens	Information Set	Ω
Cercis only (T3)	$S, \nabla S, H_S$	0 [Rigorous]
Cercis + Human (HC-i)	Above + J_H	> 0 if $\mathbb{I}(J_H; \varepsilon \mid S) > 0$
Any lens (HC-iii)	$\mathcal{I}_H \cup \text{SCX}$	0 if $\mathbb{I}(\varepsilon; \mathcal{I}_H \mid X) = 0$ [Rigorous]

Theorem 3 is not a law of physics — it is a statement about a specific lens. Change the lens, and the boundary may shift. But there exists a *limit lens* — the union of all possible observables — and at that limit, the boundary becomes absolute: **perfect noise is inseparable under any lens**.

4.1 What Makes Noise “Imperfect”?

For Theorem 3 to be breakable, noise must leave traces:

- **Annotation process traces:** time-to-label, annotator disagreement, annotator identity, revision history.
- **Perceptual traces:** blur, ambiguity, audio noise — features a human perceives but the model may not encode.
- **Semantic traces:** labels violating world knowledge (“a dog labeled ‘automobile’”), detected via commonsense.
- **Structural traces:** labels inconsistent with local manifold structure — outliers that statistical regularities miss but causal reasoning catches.

Each trace is a potential source of $\mathbb{I}(J_H; \varepsilon \mid S)$ — a wedge that human cognition uses to pry apart what the Cercis Score fuses together. Measuring each wedge’s strength is the empirical program that Theorem ?? mandates.

5 Discussion

5.1 Implications for SCX Deployment

The practical lessons form a nuanced picture:

- **Theorem 3:** Do not trust an automated noise detector that cannot state its structural assumptions — without them, its claims are unfalsifiable.
- **Theorem ??:** Do not despair — human judgment, when properly integrated, can provide the structural assumptions that Theorem 3 requires.
- **Theorem ??:** Do not expect miracles from small human budgets — human review of 1% of a million-sample dataset provides only $\sqrt{0.01} = 0.1$ of the potential gain.
- **Theorem ??:** If the noise is truly perfect, accept the limit — design systems that are *robust* to undetectable noise rather than systems that pretend to eliminate it.

5.2 The Cognitive Science Interface

Theorem ?? opens a direct interface between SCX theory and cognitive science. The key quantity — $\mathbb{I}(J_H; \varepsilon \mid S)$ — is measurable. Cognitive psychologists have studied related phenomena under *metacognitive sensitivity*

(how well can people judge their own decision accuracy?) and *confidence–accuracy calibration* [? ?]. Adapting these paradigms to the noise–difficulty distinction — presenting annotators with samples of known noise status and measuring their conditional mutual information — is the natural next experiment.

6 Conclusion

We have mapped the continuum from Theorem 3’s absolute zero ($\Omega = 0$ for AI-only), through budget-limited partial separability ($\Omega \sim \sqrt{B_H/n} \cdot \Delta_I$), to lens-dependent full separability ($\Omega > 0$ when $\mathbb{I}(J_H; \varepsilon \mid S) > 0$), bounded above by the absolute limit of perfect noise ($\Omega = 0$ when $\mathbb{I}(\varepsilon; \text{all}) = 0$). This continuum replaces a binary debate (“can AI detect noise or not?”) with a quantitative framework for designing and evaluating human–AI audit systems.

The central open problem — the empirical measurement of $\mathbb{I}(J_H; \varepsilon \mid S)$ — is a call for cognitive experiments that would determine, once and for all, whether human cognition provides a wedge into Theorem 3’s barrier, or whether the barrier is, in practice as in principle, absolute.

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